



CONTEXTUAL REPORT

Sentry: AI Powered Behavioral Analysis, Drive Health Prediction and Remote Monitoring for PC Users

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Date of submission	2026.04.25

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1.Introduction

This chapter introduces Sentry, a privacy first, local monitoring system designed to protect users from security and hardware risks. It outlines the project context, highlighting the limitations of existing tools and the need for proactive monitoring. Additionally, it details the project motivations, objectives, and technical components, specifically behavioral biometrics and SMART data analysis, before concluding with a structural roadmap of the report.

1.1. Project Background

1.1.1.Motivation

Modern computing environments introduce subtle vulnerabilities that standard security software often overlook. These risks are not only technical, but also behavioral and physical. For example, a person's use of their computer can show early signs of health problems, such as fatigue or cognitive impairment (GOWDA and Sessaiah Merikapud, 2025). These behavioral changes can include small tremors in mouse movements or prolonged keystrokes (dwell time), which can be early signs of neurodegenerative diseases such as Parkinson's or Alzheimer's (Tripathi et al., 2024). And as remote work and online learning increase, monitoring these changes is essential, but invasive 'bossware' systems completely ignore them (Munn, 2024). Beyond security and health, hardware reliability remains a major problem. SSD and hard drive failures can instantly erase years of data. Until the hardware is almost completely unrecoverable, everyday users may not notice signs of failing drives, such as slow file access or frequent crashes (Meta Research, 2023).

1.1.2.Research Problem

The market currently lacks a single solution for these overlapping issues. Passwords alone is not enough, once credentials are stolen or bypassed, an intruder has complete control over the device. Behavioral shifts driven by health problems or unauthorized access expose everyone to risk (GOWDA and Sessaiah Merikapud, 2025). Users often view existing monitoring software that uses behavioral biometrics as spyware or bossware since privacy is rarely the priority (Munn, 2024). These tools are usually for large enterprise usage and depend heavily on cloud processing, and are not designed for everyday users.

Sentry is designed to address this by operating strictly locally. It learns a user's typical typing and mouse dynamics (behavioral biometrics) to create a normal baseline, then flags unusual

patterns that may indicate cognitive fatigue, stress, underlying health issues or unauthorized access (Chitta, 2024). At the same time, it monitors SMART data and warns users about storage drive failures before their data is lost (Tomer et al., 2021). To keep a human in the loop, Sentry sends alerts via the Telegram API so that a trusted supervisor can assist remotely. All data is encrypted and processed only on the machine, proving that true security doesn't require sacrificing personal privacy.

1.2. Project Aim

To develop an AI powered monitoring system that helps monitor, remote workers, students or vulnerable users by detecting behavioral anomalies that indicate unauthorized access, fatigue or neurodegenerative conditions. And detecting SSD, hard drive failures while preserving privacy.

1.3. Objectives

- To develop an anomaly detection model that learns the user's behavioral biometrics and identifies deviations indicating unauthorized access, fatigue or health related issues, including early neurodegenerative signs like micro tremors.
- To implement storage drive health prediction using SMART attribute monitoring that predicts drive failures and warns users.
- To create a secure Telegram based alert system that notifies supervisors of potential issues while requiring user approval for remote assistance.
- To establish privacy features including local processing and encryption, a visible system tray icon, and a shutdown button that instantly stops all monitoring.

1.4. Description of the Artefact

Sentry is a locally run, privacy preserving AI monitoring system. It acts as an early warning system for both user well being and hardware integrity. The artifact will be developed using Python, utilizing it's pynput library to capture behavioral data, pySMART for drive monitoring, and Isolation Forest algorithm for machine learning.

Features of the project:

- Behavioral anomaly detection with anomaly scores.
- SMART attribute monitoring for drive health prediction.
- Real time alerts via the Telegram API.
- Remote system assistance and management.
- Privacy first architecture with locally encrypted data.
- A shutdown button to instantly stop all monitoring.

1.4.1. Project Scope

In Scope:

- **Behavioral anomaly detection:** Tracks keystroke and mouse dynamics to build a user baseline, identifying deviations from it.
- **Drive health monitoring:** Predicts SSD/HDD failures using SMART attributes.
- **Local processing:** Runs an Isolation Forest model entirely on-device, keeping all data locally encrypted.
- **Alerting and remote assistance:** Uses the Telegram API for real time supervisor notifications and consent based remote command execution.

Out of Scope:

- Any external data processing, model training, or cloud storage.
- Classifying the cause of the anomalies (e.g., distinguishing between unauthorized access and health issues)
- Large scale corporate system monitoring.
- Facial or voice recognition via webcams or microphones.
- Smartphone or smartwatch applications (the system is exclusively for PC).

1.4.2. How The Problem Will Be Solved

The system fixes the problem of subtle, undetected vulnerabilities by setting a localized baseline for normal user behavior and hardware performance. Sentry can accurately flag deviations without using invasive, cloud based processing by constantly comparing real time interactions to this normal baseline using an unsupervised machine learning model called Isolation Forest. The system sends an immediate Telegram alert to a trusted supervisor when it flags a behavioral anomaly or hardware risk. This solves the previously mentioned problems. This makes it possible to respond safely with a human in the loop, where remote help can only be given after getting the user's consent.

1.5. Report Structure

The first chapter covers the project background, aims, objectives, and a description of the artifact. Chapter 2 presents the literature review, market research, similar systems, functional and non functional requirements, and a comparative analysis. Chapter 3 details the project timeline using an Agile based Gantt chart and Work Breakdown Structure. The final chapter outlines the artifact design, planning, testing, and evaluation strategies.

2.Literature Review and Market Research

2.1. Literature Review

2.1.1.Introduction

This literature review examines the intersection of behavioral biometrics, machine learning, and hardware reliability. By evaluating current methods for anomaly detection and hardware monitoring, it identifies specific research gaps to validate Sentry's design as a comprehensive, privacy focused security system.

2.1.2.Behavioral Biometrics

In this research, behavioral biometrics deals with the continuous analysis of users' interaction with the input devices. Sentry generates a behavioral profile for the specific user with a combination of keystroke and mouse dynamics that forms an identity profile thus making it extremely difficult to fake. This multi modal fingerprint allows for the detection of unintended patterns through unobtrusive, passive monitoring (Holdsworth and Kosinski, 2025).

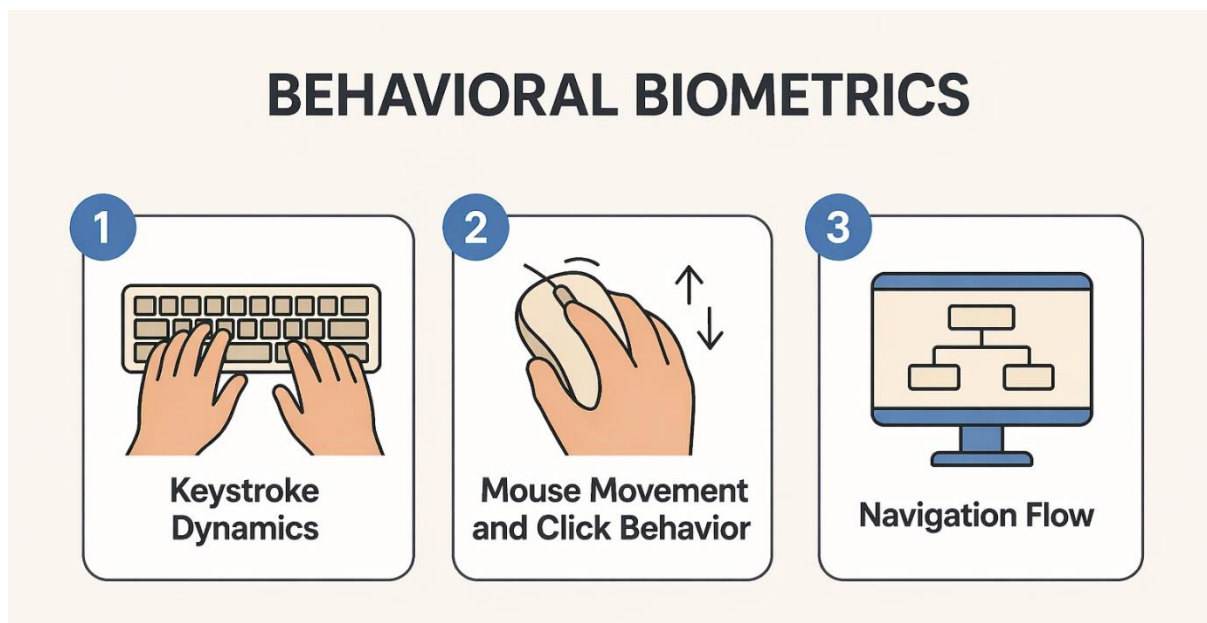


Figure 1: Behavioral Biometrics

Keystroke Dynamics

This system does not track individual keys and specific characters, but rather the time patterns of a person typing, by examining Dwell Time, Flight Time and Latency (Xi et al., 2023). The model then looks for deviations in user behavior by monitoring the small, acute intervals between two adjacent key presses, a method which does not transmit any sensitive data. It is a metadata driven approach that secures users, while still protecting privacy (Shadman et al., 2025).

Key Features (metrics):

- **Dwell Time:** The duration a key is held down
- **Flight Time:** The interval between releasing one key and pressing the next
- **Latency:** The total time between two consecutive key presses

Mouse Movements Dynamics

Mouse movements are often more descriptive than keystrokes, as users tend to maintain a consistent flow when navigating their systems (Khan et al., 2024).

- **Movement flow:** The system keeps track of speed and path curvature to create a unique behavioral signature. Most users move in smooth, predictable arcs that stay the same during normal use.
- **Behavioral Indicators:** Deviations from normal behavior such as jagged or jerky movements can signal unauthorized access, but they are also indicators of health changes, such as hand tremors or which maybe an early sign of neurodegenerative conditions.

Machine Learning Approach

The system applies an Isolation Forest algorithm to combined keystroke and mouse dynamics to detect real time baseline deviations. It analyzes typing metrics, Flight Time, Latency, and Dwell Time. Where high temporal variance may signals stress or confusion. Simultaneously, it evaluates mouse speed and path curvature. While smooth arcs indicate normal navigation, sudden jagged movements, hand tremors, or prolonged hovering indicates sign of fatigue (Pepa et al., 2021), neurodegenerative conditions or unauthorized access.

System Limitations: Detection vs Classification

Sentry performs anomaly detection rather than classification. It accurately identifies and sends an alert when the user's behavior deviates from their established normal baseline. However, the system does not explicitly classify the cause of the deviation, like distinguishing between an unauthorized intruder or health issues like hand tremors. Sentry functions as an early warning system. It alerts the designated supervisor, relying on the 'human-in-the-loop' approach to interpret the context and determine the actual cause of the anomaly.

2.1.3. Drive Health Prediction

Traditional drive maintenance is reactive, with users replacing a drive only after it fails and data is lost. Sentry shifts this approach to proactive monitoring by identifying early warning signs of failure by monitoring the SMART attributes of storage devices (Horizon Editorial, 2025).

SMART Attributes

Self-Monitoring, Analysis, and Reporting Technology (SMART) is built into most HDDs and SSDs. It continuously tracks critical health indicators:

- **SMART 5 (Reallocated Sectors):** Remapped bad sectors. A rising count predicts failure.
- **SMART 187 (Reported Uncorrectable Errors):** Errors the drive couldn't fix.
- **SMART 188 (Command Timeout):** Aborted operations due to unresponsive drive.
- **SMART 197 (Current Pending Sectors):** Unstable sectors waiting to be remapped.
- **SMART 198 (Uncorrectable Sectors):** Total uncorrectable read/write errors.
- **SMART 194 (Temperature)**

These attributes reveal the drive's health status in real time. If any of these has a sudden spike, it means the drive is starting to degrade.

The Problem with Traditional Monitoring

Traditional systems rely on threshold based BIOS alerts that trigger too late for data recovery. Users rarely check SMART data logs, they usually only notice issues late through symptoms like slow file access and crashes right before complete failure.

Machine Learning Approach

Instead of waiting for critical thresholds, machine learning (Isolation forest) is applied to these historical SMART data to identify patterns that precede future drive failure.

2.1.4. Machine Learning Algorithm: Isolation Forest

This system uses the Isolation Forest algorithm (unsupervised learning) for all AI components. It trains on initial data to establish a normal baseline, which can be easily retrained if necessary. It then finds outliers by separating data points that are rare or strange. Anomalies diverge more quickly from the baseline because there are fewer of them and they are more unique (Liu, Ting and Zhou, 2008).

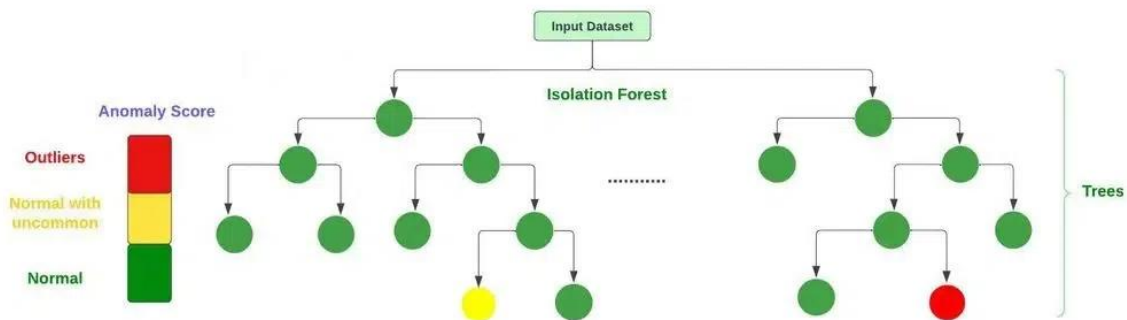


Figure 2: Isolation Forest

Why Isolation Forest:

Supervised algorithms, like random forest, need labeled datasets to figure out what is normal. There are generalized biometric datasets, but they only show how people act on average, not how one certain person acts. When this general model is used, it often gives false positives when the user's natural patterns don't match the broader average.

In the same way, each storage device has its own wear patterns and usage history. A model trained on aggregate SMART data cannot accurately track the degradation of one specific drive. What is considered an anomaly for one drive may be normal for another.

Isolation Forest fixes both problems by using the user's behavioral data and the drive's data to set baselines for one specific user or device. This unsupervised algorithm makes a very personalized profile during its first training period by not using external general datasets.

2.1.5.Alerting Remote Assistance

Sentry enables authorized supervisors to receive real time alerts via the Telegram API when behavioral anomalies or drive health risks are detected. These notifications allow supervisors to assess the issue and cause and provide remote assistance if needed, which is executed only with the user's explicit consent.

Remote Assistance

This operates through a command based system. Supervisors send commands via Telegram, which are received and parsed by Sentry. When a command is received, the system:

1. Validates the command
2. Presents a consent prompt to the user showing what action is requested
3. Executes the action after user approval

With user consent, authorized supervisors can execute specific remote actions. Examples include blocking access to malicious websites, opening applications, terminating suspicious processes and more.

2.1.6.Privacy and Control

Sentry is designed with a privacy first, locally run architecture, avoiding the cloud based processing like existing tools (Sokhansanj, 2025).

- **Local processing:** All biometric and hardware data is encrypted and stored exclusively on the device, no sensitive data ever leaves the device.
- **Consent based action:** Remote assistance require explicit user approval for command execution.
- **Transparency:** A persistent system tray icon ensures the user is always aware of active monitoring.
- **Shutdown button:** A shutdown button allows users to instantly terminate all monitoring at any time.

2.1.7.Existing Systems

Behavioral Biometric Systems:

- **URIEL (KIT Analysis):** This system identifies users by mapping unique typing rhythms via Key Interval Time (KIT). By splitting the keyboard into groups and recording a 10x10 matrix of transition times, it trains a dedicated AI model for each user. It achieved a 93% accuracy rate using the Least Hated method, maintaining a low 2% insecure failure rate (Purgason and Hibler, 2012).
- **On-Device Behavioral Trust Engine:** A local mobile and wearable system calculating a trust score in under 50ms. It accepts the correct user >90% of the time with a <5% false acceptance rate (Chitta, 2024).
- **LSTM Identity Verification:** This approach uses Long Short-Term Memory (LSTM) networks to continuously authenticate remote workers based on the flow of keystrokes and mouse movements. LSTMs were chosen for their ability to learn behavioral rhythms over time, achieving 89% accuracy in distinguishing legitimate users from imposters (Awawdeh et al., 2024).
- **Multi-modal Continuous Authentication:** Combines typing, mouse, and smartphone motion tracking. Using a hybrid GRU and CNN architecture, it reached 97.2% accuracy (Ejaz, 2025).

Drive Health Prediction Models:

- **Decision Trees & GBRT:** These models analyze SMART attributes like temperature and reallocated sectors to predict failures. While Decision Trees provide clear, interpretable failure classifications, Gradient Boosted Regression Trees (GBRT) offer failure probabilities (e.g., 85%) to help prioritize maintenance. These models detected over 93% of failures with a negligible 0.01% false alarm rate (Li et al., 2017).
- **Random Forest (SMART Attribute Analysis):** Trained on large Backblaze datasets to replace basic threshold alerts. It isolated five critical SMART indicators, proving Random Forest works best for balancing successful detections against false positives (Tomer et al., 2021).
- **Classification and Regression Trees (CART):** This research critiques binary fail/no-fail models, proposing a Health Degree score to track gradual drive deterioration. The Classification Tree (CT) caught 95% of failures, and implementing these scores in RAID-6 setups significantly improved overall system reliability and data safety (Li et al., 2014).

2.1.8. Research Gap

Despite the technical strengths of the systems discussed above, several critical gaps remain that Sentry is designed to address:

- **Fragmentation:** Behavioral detection, drive health, and remote assistance exist only as separate systems.
- **Privacy:** Most AI tools rely on cloud processing, forcing users to send sensitive data to external servers.
- **Consumer focus:** Behavioral tools are built for corporate surveillance, not everyday users.

Sentry addresses these by combining behavioral analysis, drive health monitoring, and remote assistance in a locally run platform that keeps all data on the user's device for strict privacy. Unlike corporate monitoring tools, Sentry focuses on consumer security with transparent, non-intrusive monitoring that prioritizes individual well-being and data privacy.

Table 1: Similar systems and proposed system feature comparison

Feature	On-Device Behavioral Trust Engine	URIEL (KIT Analysis)	Sentry
Behavioral Analysis	Yes	Yes	Yes
Drive Health Monitoring	No	No	Yes
Remote Assistance	No	No	Yes
Mobile Support	Yes	No	No
Wearable Support	Yes	No	No
Specific Keyboard Grouping	No	Yes	No

2.2. Market Research

2.2.1. Market Research Planning

The market research was designed to validate Sentry's features. A Google Forms survey was targeted at potential supervisors such as parents, managers, teachers, and IT admins. A sample size of 100 respondents was selected to simplify percentage calculations, reveal clear trends, and fit within the project timeframe. A convenience sampling method was used to quickly reach potential users and evaluate features. The questionnaire assessed these main aspects:

- User demographics and intended monitoring users
- Concerns regarding user fatigue, health, and unauthorized access
- Expectations for privacy
- Past data loss, backup habits, and the need for drive health alerts
- Comfort levels with using Telegram for alerts and remote assistance
- Potential barriers to adoption and willingness to install the system

This targeted approach ensured rapid data collection, captured real world privacy expectations, and directly guided the final technical requirements.

2.2.2. Data Collection Methods

Respondents answered 15 questions in a Google Forms survey. The questionnaire focused on privacy expectations, backup habits, behavior tracking, and remote Telegram support. While this method enabled rapid data collection, reliance on these specific sampling techniques introduces potential bias that limits the ability to make broad generalizations.

2.2.3. Survey Data Visualization and Analysis

What is your age group?
100 responses

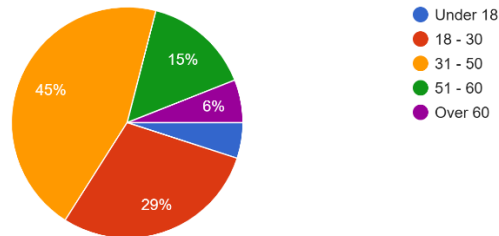


Figure 3: Question 1 Age group

Figure 3 shows the majority of respondents belong to the 31-50 age group (45%), followed closely by the 18-30 group (29%). This indicates that the primary target audience configuring and deploying the system consists of younger to middle aged adults.

If you were to use a system like Sentry, who would you most likely be monitoring?
100 responses

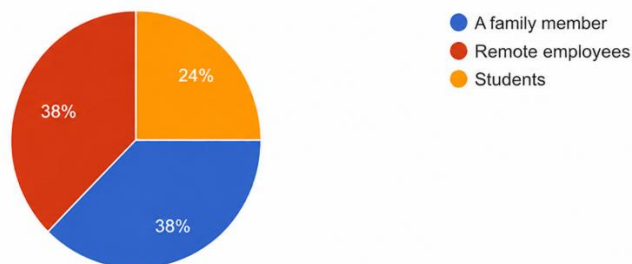


Figure 4: Question 2 Monitoring target

Figure 4 shows 38% of respondents each would primarily use the system to monitor remote workers and family members, while 24% would use it to monitor students. This validates the system's dual use case for both personal health support and corporate security.

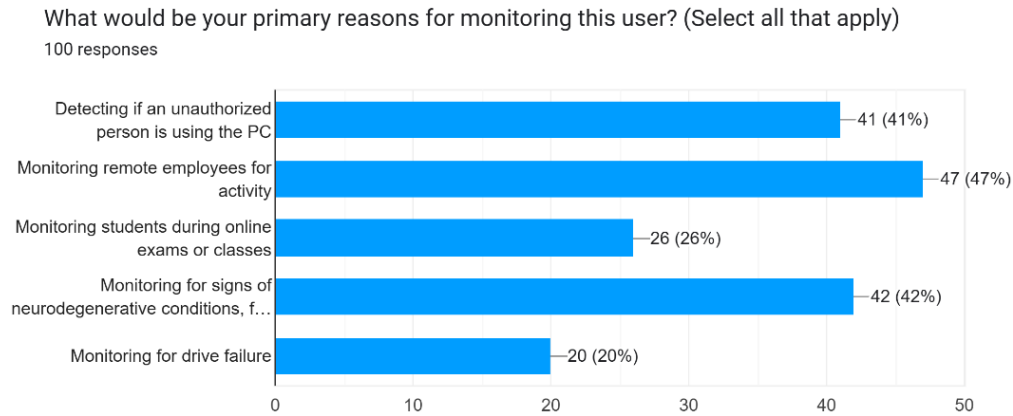


Figure 5: Question 3 Use cases

Figure 5 shows the most significant motivation for using the system is for monitoring remote employees (47%). This is followed closely by monitoring for signs of fatigue, stress, or neurodegenerative conditions (42%) and to detect unauthorized access (41%).

Have you ever wished you could check on them without being physically present or intrusive?

100 responses

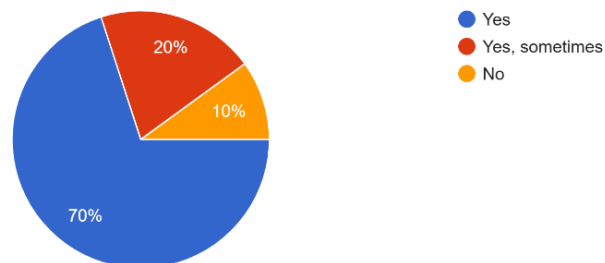


Figure 6: Question 4 Non-intrusive monitoring

Figure 6 shows 70% of respondents stated they have wanted a way to check on their target user without being physically present or intrusive, validating the privacy focused nature of the application.

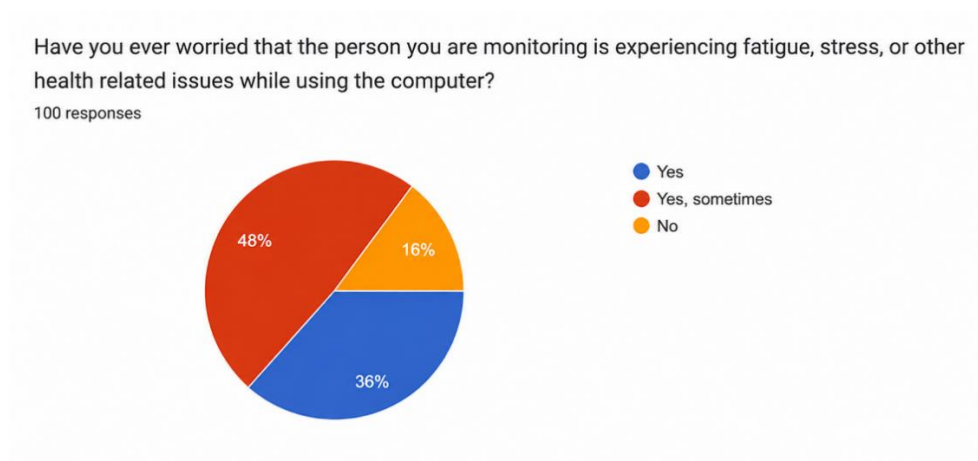


Figure 7: Question 5 Health concerns

Figure 7 shows almost 50% of respondents sometimes worry that the person they are monitoring might experience fatigue, burnout, or health issues while using the computer.

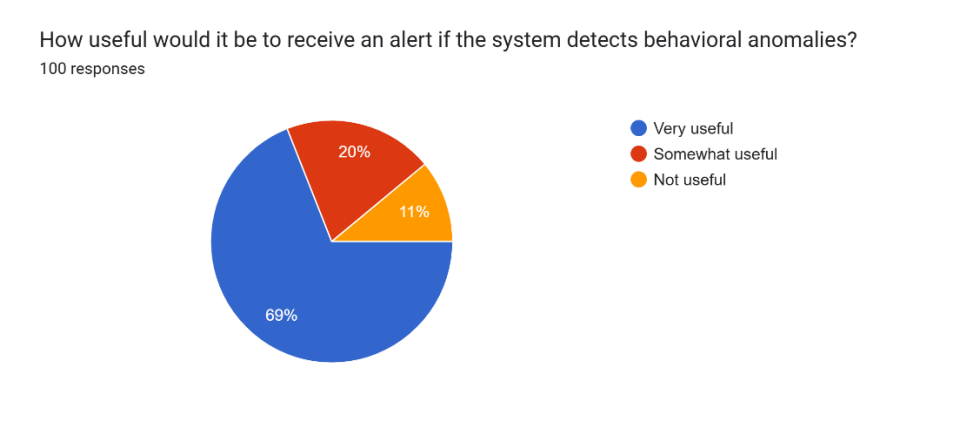


Figure 8: Question 6 Alert usefulness

Figure 8 shows 70% of participants consider receiving alerts for behavioral anomalies to be very useful, with another 20% finding it somewhat useful.

If an issue was detected, would you want the ability to remotely assist them, provided they give consent?

100 responses

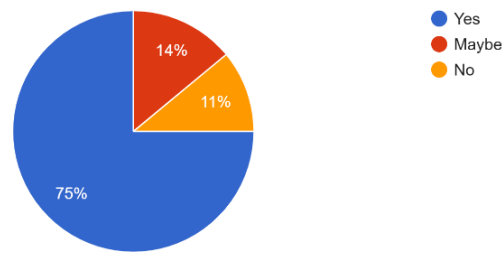


Figure 9: Question 7 Remote assistance

Figure 9 shows 75% of the participants want the ability to remotely assist the user upon receiving an alert, confirming the necessity of the Telegram remote command feature.

How concerned are you about someone gaining unauthorized access to their computer?

100 responses

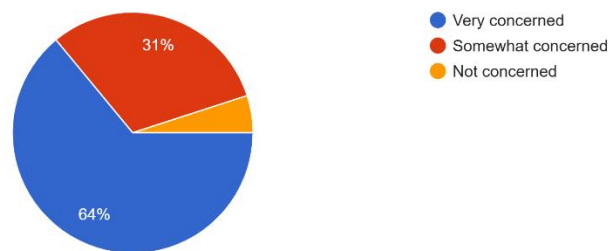


Figure 10: Question 8 Unauthorized access

Figure 10 shows 64% of the respondents expressed concern about someone gaining unauthorized access to the target computer, further validating the need for the behavioral biometric detection.

How important is the privacy of their sensitive data such as behavioral data (typing patterns, mouse dynamics)?

100 responses

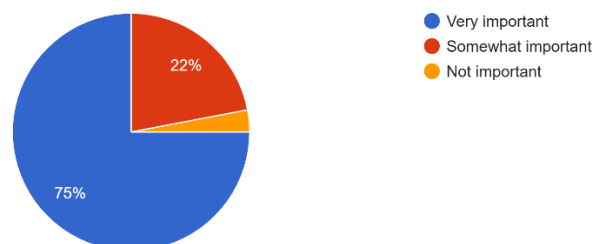


Figure 11: Question 9 Data privacy

Figure 11 shows 75% of users consider the privacy of their behavioral data to be very important. This strongly justifies the system requirement to process all data locally rather than relying on a cloud server.

Has the person you monitor ever lost important data due to an unexpected drive failure?
100 responses

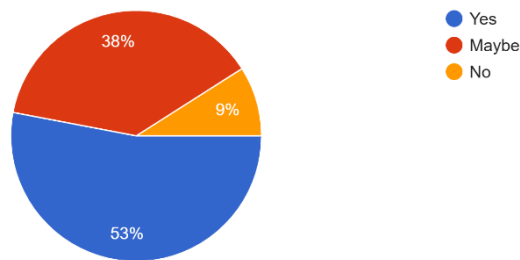


Figure 12: Question 10 Data loss

Figure 12 shows a total of 90% of the participants may have previously lost important data due to an unexpected drive failure.

How often do they back up their computer?
100 responses

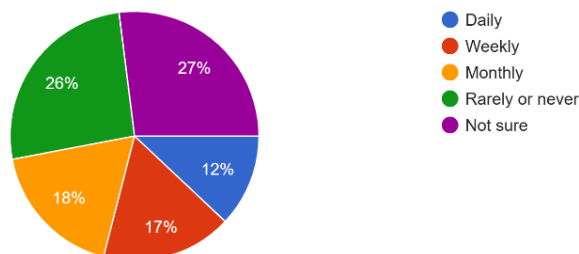


Figure 13: Question 11 Backup habits

Figure 13 shows a total of 44% of the respondents either rarely back up their devices or only do so on a monthly basis, leaving them highly vulnerable to sudden disk failures.

How useful would it be to get an early warning if their computer's storage drive was about to fail?
100 responses

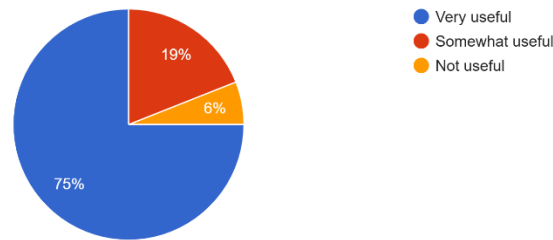


Figure 14: Question 12 Drive health alert

Figure 14 shows 75% of users would find early drive health predictions very useful. This feature directly addresses the vulnerability highlighted by their poor backup habits in the previous figure.

Sentry uses the Telegram API to send alerts to you. How comfortable are you with using Telegram to receive alerts and execute remote actions?
100 responses

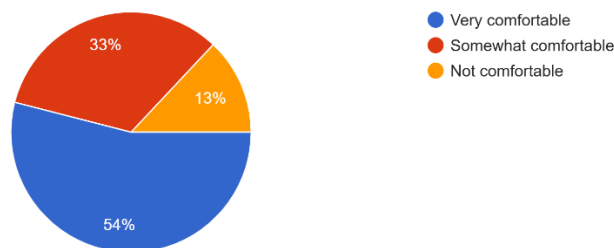


Figure 15: Question 13 Telegram usage

Figure 15 shows more than 50% of the surveyed individuals are comfortable using the Telegram application to receive alerts and execute remote actions.

What would be your concerns about using a tool like Sentry?
100 responses

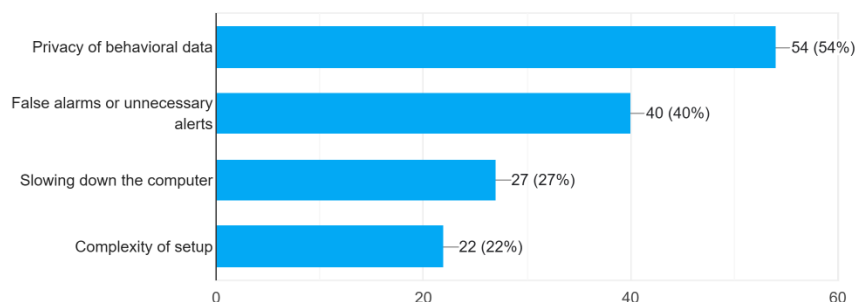


Figure 16: Question 14 Concerns about system

Figure 16 shows the biggest barriers to adoption are concerns about the privacy of personal data (54%) and the potential for false alarms (40%). This highlights the need for a highly accurate machine learning model and a transparent, localized architecture.

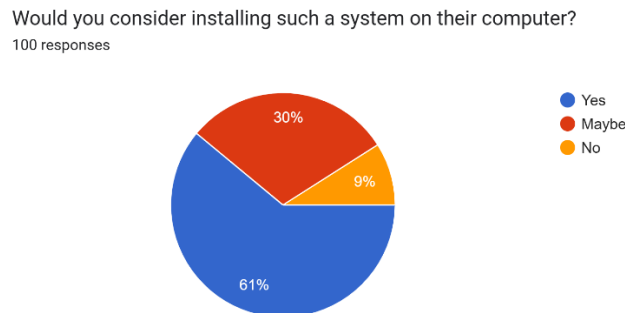


Figure 17: Question 15 Consider installation

Figure 17 shows a combined total of 91% of respondents stated they would either definitely (61%) or potentially (30%) consider installing such a system on their target's computer.

2.2.4. Conclusion

The survey received 100 responses and validates the need for a system like Sentry. Potential supervisors are concerned about unauthorized access, health related issues and hardware failures, and they also demand strong data privacy. Telegram integration with local processing architecture and consent based authentication directly addresses this issue. With 91% of respondents willing to adopt the system, the findings clearly confirm Sentry as a needed solution.

3. Project Plan

This chapter describes planning and scheduling for Sentry. Using an Agile methodology, the workload is divided into specific sprints. A WBS and Gantt chart organize tasks, some running in parallel and others running sequentially, ensuring efficient time management and timely project completion.

3.1. Gantt Chart

This Gantt chart outlines the Agile timeline for the Sentry project, tracking key milestones. Completed tasks appear in green, while pending ones are blue. Visualizing the sprints for behavioral anomaly detection and hardware monitoring ensures efficient time management.

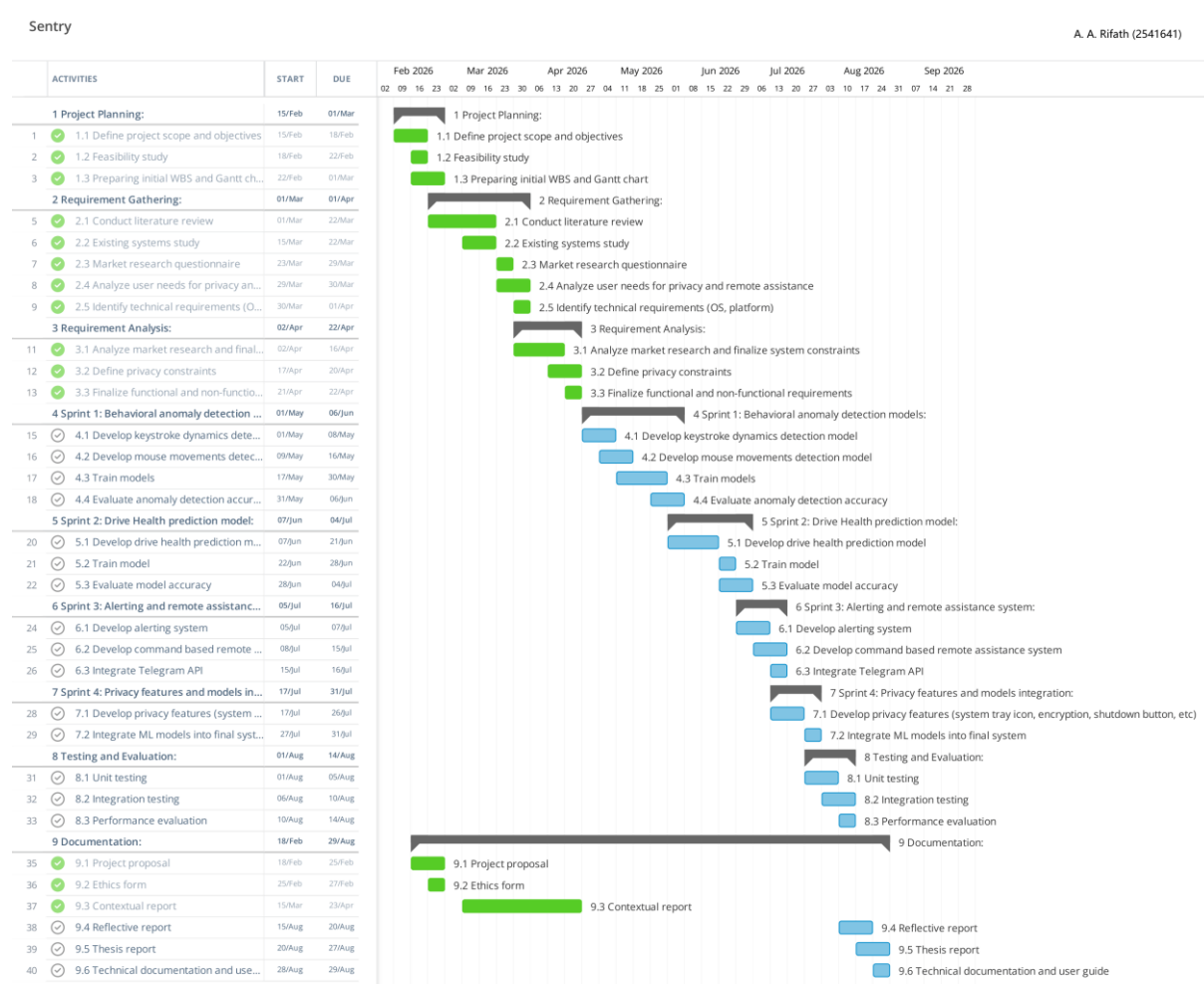


Figure 18: Gantt Chart

3.2. Work Breakdown Structure (WBS)

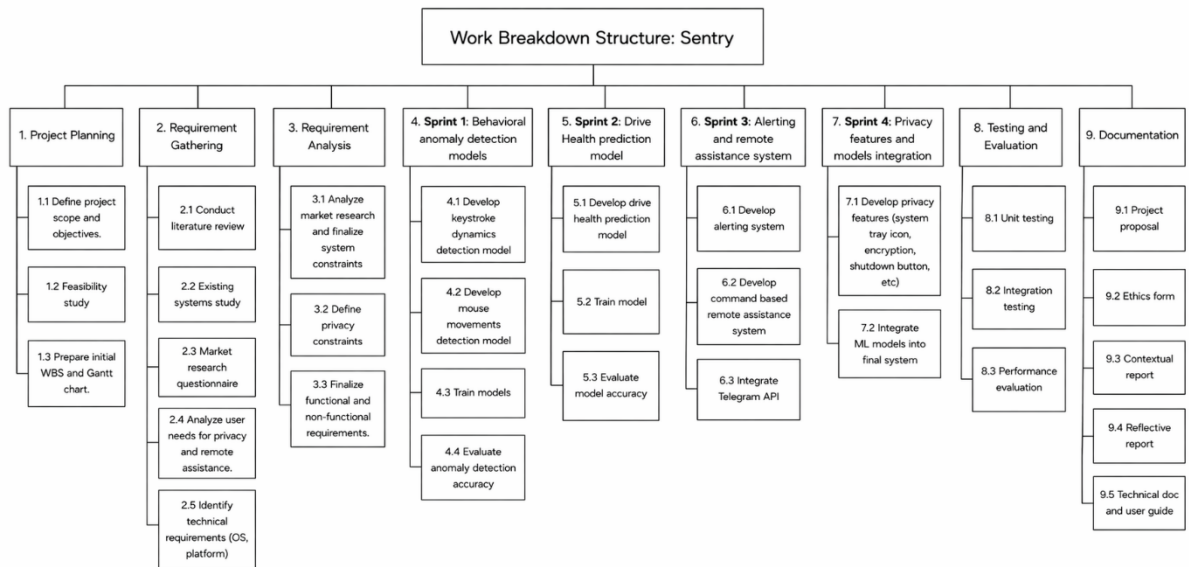


Figure 19: Work Breakdown Structure

4. Planning the Artifact

This chapter details the full development plan. About the chosen Agile methodology, requirement analysis, system architecture, implementation strategies, and testing protocols. Finally, this gives a clear and organized plan for making, testing, and designing the monitoring artifact that protects privacy.

4.1. Methodology Selected and Why

The development utilizes an Agile methodology rather than a traditional linear method such as Waterfall method for four reasons:

- **Iterative model tuning:** The Isolation Forest algorithm requires continuous training. Agile allows for regular accuracy evaluations and adjustments after every sprint.
- **Modular Integration:** Building and connecting the system's components sequentially significantly lowers the risk of major bugs.
- **Security validation:** Since local data processing is essential, Agile ensures encryption protocols are tested continuously instead of waiting until the final testing phase.
- **Time management:** Breaking the workload into manageable sprints keeps development tightly aligned with the deadline.

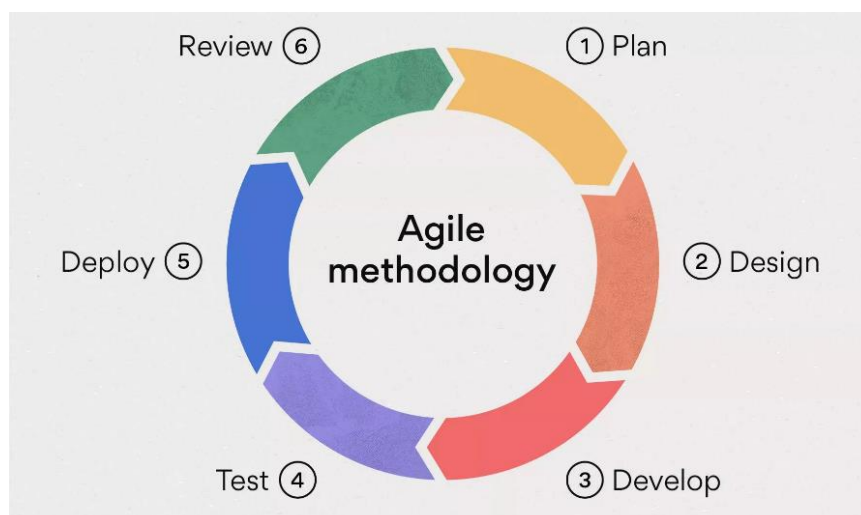


Figure 20: Agile Methodology

Table 2: Agile vs Waterfall comparison

Agile Methodology	Waterfall Methodology
Continuous Isolation Forest tuning improves accuracy.	Late adjustments to ML models are difficult.
Step by step integration catches bugs early.	End stage integration risks major system failures.
Encryption and privacy are tested every sprint.	Late testing exposes fundamental security flaws.
Delivers a working prototype early.	No functional software until the final weeks.

4.2. Planning

The planning phase defined the development, scheduling, and monitoring of the Sentry artifact. WBS and Gantt chart managed tasks and milestones across four Agile sprints:

- **Sprint 1:** Developed behavioral anomaly detection models
- **Sprint 2:** Developed drive health prediction model
- **Sprint 3:** Developed alerting and remote assistance system
- **Sprint 4:** Developed privacy features and models integration

The Gantt chart provided a clear timeline for task durations and dependencies. This structured approach ensured all research, development, and evaluation activities were completed within the allocated timeframe, reducing the risk of delays.

4.3. Requirement Gathering and Analysis

As discussed in previous chapters, data collection drove the project's technical direction. The primary data gathering method was the market research survey which captured user expectations regarding privacy and behavioral tracking. The secondary data gathering method was the literature review which exposed the security flaws in existing cloud dependent software. Together, these findings justified a localized architecture and defined the functional requirements for the Agile sprints.

Table 3: Software Requirements

Software Component	Requirements
Operating system	Windows 10+
IDE	Visual Studio Code
Programming language	Python 3.14+
Machine learning algorithm	Isolation Forest (scikit-learn)
Libraries and packages	scikit-learn, pynput, pySMART, python-telegram-bot
API	Telegram bot API
Version control	Git, Github

Table 4: Hardware Requirements

Hardware Component	Requirements
CPU	Dual core 2.0 GHz or higher. No GPU is needed since Isolation Forest is lightweight.
RAM	4GB minimum. 8GB recommended.
Storage	1 to 2GB of free space
Device	Laptop or desktop

4.3.1. Functional and Non-Functional Requirements

Functional requirements:

- The system shall capture keystroke and mouse data to establish a behavioral baseline.
- The system shall continuously compare live behavior against the baseline to flag anomalies.
- The system shall detect deviations indicating unauthorized access, fatigue, or neurodegenerative conditions.
- The system shall analyze SMART data to identify drive pre-failure patterns.
- The system shall issue warnings when drive failure thresholds are met.
- The system shall notify supervisors via Telegram upon detecting anomalies or hardware risks.
- The system shall allow supervisors to execute remote commands via Telegram.
- The system shall display a system tray icon during active monitoring.

Non-Functional requirements:

- The system should process, store and encrypt all data strictly locally.
- The system should run continuously as a background service.
- The system should automatically queue and retry failed Telegram alerts.
- The system should require explicit consent before executing remote actions.
- The system should maintain a local audit log of all commands and alerts.
- The system should operate with minimal CPU and RAM usage.

4.4. Design

Sentry's privacy first architecture will rely entirely on local processing, operating as a background service without cloud connectivity. The system will comprise four interconnected modules:

- **Behavioral Logging:** Will capture raw keystroke and mouse dynamics using pynput library.
- **Drive Health:** Will extract SMART attributes via pySMART.
- **Machine Learning:** Will process behavioral and hardware data locally.
- **Alerting and assistance:** Will manage secure alerts and remote commands via the Telegram API.

These components will communicate internally. External network interaction will occur solely when alerting pushes warning notifications to a supervisor's Telegram and during remote assistance.

4.5. Implementation

Following Agile principles, the development will proceed through these key steps:

- The system will securely log and encrypt mouse and keystroke inputs to establish a normal behavioral baseline.
- An unsupervised Isolation Forest model will train locally using scikit-learn. And can be easily retrained as needed to maintain accuracy.
- The final phase will connect the Telegram bot, setting up webhooks for remote commands and on-screen consent prompts.

4.6. Testing and Evaluation

This phase will ensure Sentry operates accurately while maintaining strict privacy constraints:

Testing:

- **Unit & Integration testing:** Verify individual modules (pynput, pySMART) and secure Telegram data flows.
- **ML accuracy:** Simulate typing anomalies to balance the Isolation Forest model FAR, FRR, and EER.

Evaluation:

- **Performance:** Track CPU/RAM usage over 24 hours to ensure the background service is efficient.
- **Scenario:** Trigger fake anomalies to verify Telegram alert speeds and remote command execution.

4.7. Risks and Mitigation techniques

Table 5: Risks and Mitigation techniques

Risk	Impact	Probability	Mitigation
Inaccuracy in behavioural anomaly detection	High	High	Tune model thresholds and improve training data.
Privacy concerns	Medium	Low	Ensure transparency, user consent, encryption, and shutdown option.
Telegram API service disruption	High	Low	Implement alternate alerting method until service restores.
Inaccuracy in drive health prediction	High	Medium	Tune model thresholds and improve training data.
Time constraints	Medium	Low	Prioritize core features before optional enhancements.

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6. Appendices

Appendix A: Questionnaire Questions

Link: <https://forms.gle/m3wGe6NLmzWoLJBMA>



Sentry: AI Powered Behavioral Analysis, Drive Health Prediction and Remote Monitoring for PC Users

This survey is part of my final year computer science research project at University of Bedfordshire evaluating Sentry, a privacy-focused, AI-powered monitoring system. Sentry analyzes keystroke and mouse dynamics locally on the device to detect behavioral anomalies linked to health related issues like fatigue, stress, early signs of neurodegenerative conditions. Or unauthorized access. It also predicts storage drive failures using SMART data to prevent sudden data loss. If a risk is detected, Sentry alerts a trusted overseer/supervisor who can provide remote assistance if needed.

Please take 3-5 minutes to share your thoughts on whether this tool would be useful in the real world. Thank you for your support!

What is your age group? *

- ☐ Under 18
- ☐ 18 - 30
- ☐ 31 - 50
- ☐ 51 - 60
- ☐ Over 60

On a scale of 1 to 5, how comfortable are you with using a computer? *
(1 = Beginner, 5 = Expert)

- | | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| ☆ | ☆ | ☆ | ☆ | ☆ |

If you were to use a system like Sentry, who would you most likely be monitoring? *

- ☐ A family member
- ☐ Remote employees
- ☐ Students

What would be your primary reasons for monitoring this user? (Select all that apply) *

- ☐ Detecting if an unauthorized person is using the PC
- ☐ Monitoring remote employees for activity
- ☐ Monitoring students during online exams or classes
- ☐ Monitoring for signs of neurodegenerative conditions, fatigue or stress
- ☐ Monitoring for drive failure

Have you ever wished you could check on them without being physically present or intrusive? *

- ☐ Yes
- ☐ Yes, sometimes
- ☐ No

Have you ever worried that the person you are monitoring is experiencing fatigue, stress, or other health related issues while using the computer? *

- ☐ Yes
- ☐ Yes, sometimes
- ☐ No

How useful would it be to receive an alert if the system detects behavioral anomalies? *

- ☐ Very useful
- ☐ Somewhat useful
- ☐ Not useful

If an issue was detected, would you want the ability to remotely assist them, provided they give consent? *

- ☐ Yes
- ☐ Maybe
- ☐ No

How concerned are you about someone gaining unauthorized access to their computer? *

- ☐ Very concerned
- ☐ Somewhat concerned
- ☐ Not concerned

How important is the privacy of their sensitive data such as behavioral data (typing patterns, mouse dynamics)? *

- ☐ Very important
- ☐ Somewhat important
- ☐ Not important

Has the person you monitor ever lost important data due to an unexpected drive failure? *

- ☐ Yes
- ☐ Maybe
- ☐ No

How useful would it be to get an early warning if their computer's storage drive was about to fail? *

- ☐ Very useful
- ☐ Somewhat useful
- ☐ Not useful

How often do they back up their computer? *

- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Rarely or never
- ☐ Not sure

Sentry uses the Telegram API to send alerts to you. How comfortable are you with using Telegram to receive alerts and execute remote actions? *

- ☐ Very comfortable
- ☐ Somewhat comfortable
- ☐ Not comfortable

What would be your concerns about using a tool like Sentry? *

- ☐ Privacy of behavioral data
- ☐ False alarms or unnecessary alerts
- ☐ Slowing down the computer
- ☐ Complexity of setup

Would you consider installing such a system on their computer? *

- ☐ Yes
- ☐ Maybe
- ☐ No